

Discussion on Backend Development

17-316/616 Fall 2025

AI Tools for Software Development

<https://ai-developer-tools.github.io>

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Reflection prompts this week

1. Let's say you have already built your frontend with a mocked up backend, but now that you are building the backend, you have decided to change its external APIs from the mocked ones. How difficult is it to change the frontend to match? Discuss 3 prompts you have found to be useful to perform this task and explain how they helped.
2. Discuss the challenges of getting the LLM to create a multi-part back end design or implementation? Describe the difficulties in ensuring 1) correctness of each part, 2) consistency across the parts, and 3) accuracy of the results matching your intentions. Tell us how you would test, validate, and refine the outputs.
3. How should an LLM enable you and a colleague to collaborate on the design of a full stack application without both of you typing into the same computer prompt? One you focuses on the front end and the other specializes in the backend. Speculate on your ideal collaborative system design.
4. Use the LLM to simulate the personas of software engineering roles on your team. Define the following 3 important engineering roles: database administrator, site reliability engineer, release engineer. Provide a good LLM prompt for each that enable them to accomplish their specialized engineering task on your application's codebase. Try this out on your project's application and report how well they worked.
5. Could an LLM reason about tradeoffs for the use of external backend services to power your app? 1) Write an LLM prompt to make an argument about performance to your lead developer to adopt a particular service out of a choice of three. 2) Write another LLM prompt to make an argument about the budget to your chief financial officer to adopt a particular service out of a choice of three. 3) Why do you believe these prompts will produce reasonable analyses?
6. You are the CEO of a company who designs self-driving car software that is very popular and is sold to many car manufacturers. Your "auto-pilot" algorithm, which was designed and implemented completely by agentic AI LLMs, was in control of a car that ran down and killed a pedestrian. How do you defend your company against liability for this tragic occurrence? What evidence could an LLM help you supply in court to make your case? Give 3 LLM prompts to provide the needed exculpatory evidence.

Discussion Instructions

- Instructor announces the reflection question.
- If you chose this reflection question to answer for homework, come up to the front. You are the discussant group.
- Class forms into pairs.
- Pairs and discussant group should now reflect on the question by themselves (4 min)
 - Explain what you think, why, and how you've come to your conclusions.
- Each discussant explains their answer to the class (3-4 min)
- Instructor will choose a pair to report their answer to the class (1 min)
- A discussant will respond and connect the answer to their own reflection (1 min)

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Next Class

- P4: Backend Development due 11:59pm tonight.
- Work through the challenges of getting an LLM to generate a dev spec. Rework P2 dev specs.